

# Pietro Mesquita Piccione

## Résumé

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### Work experience

2025 – 2027 **Postdoc**, *University of Gothenburg*, Göteborg, Sweden, Supervisor: David Witt Nyström

### Education

2022 – 2025 **PhD in Mathematics**, *Institut des mathématiques de Jussieu*, Sorbonne Université, Paris, France  
Advisors: Sébastien Boucksom and Tat Dat Tô

2021 – 2022 **Master's Degree in Mathematics**, *Institut des mathématiques de Jussieu*, Sorbonne Université, Paris, France

2017 – 2021 **Bachelor's degree in Mathematics**, *Universidade de São Paulo*, USP, São Paulo, Brazil  
Prize for outstanding performance.

### Research projects and grants

2026 – 2027 **Research grant: "Kähler and non-Archimedean geometry"**, *Funded by Stiftelsen Wilhelm och Martina Lundgrens Vetenskapsfond*  
Grant of 50.000 SEK for research purposes.

2022 – 2025 **MathInParis2020**, *Scholarship for a PhD in Paris*, Cofunded by Marie Skłodowska-Curie Actions, Fondation sciences mathématiques de Paris

2021 – 2022 **PGSM Scholarship**, *Scholarship for a Master's degree in Paris*, Fondation sciences mathématiques de Paris

2020 – 2021 **Morse Theory**, *Undergraduate Research Project*, FAPESP  
Grant FAPESP 2020/04871-0

### Articles

2025 **A transcendental non-Archimedean Calabi–Yau Theorem with applications to the cscK problem**, *joint with David Witt Nyström*, preprint, arXiv:2509.09442

Description: We solve the non-Archimedean Monge–Ampère equation for compact Kähler manifolds, generalizing a result of Boucksom–Favre–Jonsson to the transcendental setting. This allows us to derive a valuative criterion for uniform K-stability for models. As a central application, we prove that this stability condition implies the existence of a unique constant scalar curvature Kähler metric.

2024 **A non-Archimedean theory of complex spaces and the csc-K problem**, *published*, *Advances in Mathematics*

Description: In this paper I construct a space of valuations attached to a compact complex manifold, that generalize the Berkovich analytification of a projective variety over  $\mathbb{C}$ . As an application I get a sufficient algebraic condition for the existence of constant scalar curvature Kähler metrics on a given Kähler class on a Kähler manifold.

### Professional

#### Teaching

2026 **Riemann Surfaces**, *PhD course*, U. of Gothenburg and Chalmers University of Technology, Sweden

#### Teaching Assistantships

2025 **Introduction to Riemann Surfaces**, *Master's course*, Sorbonne Université, France

2024 **Topology and Differentiable Calculus I**, *Second year bachelor's course*, Sorbonne Université, France

2024 **Functional Analysis**, *Third year bachelor's course*, Sorbonne Université, France

2023 **Topology and Differentiable Calculus I**, *Second year bachelor's course*, Sorbonne Université, France

- 2020 **Introduction to Algebraic Topology**, *Fourth year bachelor's course*, Universidade de São Paulo, Brazil
- 2019 **Introduction to Analysis**, *Third year teacher's diploma course*, Universidade de São Paulo, Brazil
- 2019 **Differentiable functions and series**, *Second year Computer Science's course*, Universidade de São Paulo, Brazil
- 2018 **Calculus II**, *First year Engineering course*, Universidade de São Paulo, Brazil

#### Scientific Visits

- 2025 **Simons Semester in Analysis and Geometry on Complex Manifolds**, *Erdős Center*, Budapest, Hungary  
Three months visit to the program.
- 2024 **Special Geometric Structures and Analysis**, *SLMath*, Berkeley  
Two months visit to the program.

#### Academic Services

- 2025 **Participated at the cycle "Un texte, une aventure mathématique"**, *Lycée international St Germain en Laye*, Slides and material  
Description: Gave a "pre-talk" at a high school preparing for the talk "Dennis Sullivan, mathématicien des analogies" given by Barbara Schapira. "Un texte, une aventure mathématique" is a joint organization of the Société Mathématique de France and the Bibliothèque nationale de France to expose the general public with the history and problems of current research Mathematics.
- 2021 **Organizer of the A5 Group**, Instituto de Matemática e Estatística, Universidade de São Paulo  
A student runned academic group with over 25 weekly seminars organized
- 2018 – 2019 **Deputy Student Representative in the Mathematics Department Council**, Instituto de Matemática e Estatística, Universidade de São Paulo, Brazil

#### Events Organized

- 2026 **Geometry and Topology of (Almost) Complex Structures**, *Online conference*
- 2022 **MaThematical Days : Paul Erdős**, *Universidade de São Paulo*, São Paulo, Brazil
- 2021 **MaThematical Days : Emmy Noether**, *Universidade de São Paulo*, São Paulo, Brazil
- 2021 **MaThematical Days : Élie Cartan**, *Universidade de São Paulo*, São Paulo, Brazil

### Talks

#### $\widehat{K}$ -stability on Kähler manifolds

- 2026 **Workshop on K-stability**, East China Normal University, Shanghai, China
- The non-Archimedean approach of the Yau–Tian–Donaldson conjecture**
- 2026 **Geometry, HOlonomy and Structures with Torsion**, University of Münster, Münster, Germany
- 2026 **Geometric analysis seminar**, Westlake University, Hangzhou, China
- 2026 **Geometry and topology seminar series**, University of Science and Technology of China, Hefei, China
- 2025 **Séminaire de géométrie**, Centre de mathématiques Laurent-Schwartz, Palaiseau, France
- 2025 **Workshop on Singular canonical Kähler metrics on compact and non-compact manifolds**, Rényi Institute, Budapest, Hungary
- 2025 **Geometry seminar**, Max Plank Institute for Mathematics in the Sciences, Leipzig, Germany
- 2025 **Complex geometry seminar**, Mathematisches Institut, University of Münster, Münster, Germany
- 2025 **MARGE in Brest: Fibrations and Deformations**, Université de Bretagne Occidentale, Brest, France
- 2025 **Geometry Webinar AmSur /AmSul**, , Webinar
- 2025 **Séminaire de Géométrie complexe**, Université Toulouse III - Paul Sabatier, Toulouse, France
- 2025 **Séminaire Géométries ICJ**, Université Claude Bernard Lyon 1, Lyon, France
- 2025 **Mathematics Seminar**, Aarhus University, Aarhus, Denmark
- 2024 **Séminaire de géométrie algébrique de Jussieu**, ENS, Paris, France
- 2024 **Séminaire systèmes dynamiques et géométrie**, Université d'Angers, Angers, France
- 2024 **Complex Analysis Seminar**, University of Bayreuth, Bayreuth, Germany

- 2024 **Geometric Analysis Seminar**, BICMR, Beijing, China
- 2024 **KASS Seminar**, Chalmers University, Gothenburg, Sweden
- 2024 **Algebraic Geometry Seminar**, University of Glasgow, Scotland
- 2024 **Any Complex Geometry Seminar**, University of California Berkeley, US
- 2024 **Graduate Student Seminar Series**, SLMath, Berkeley, US

#### A counterexample for the periodic orbit conjecture

- 2022 **Groupe de Travail : Dynamiques Sauvages**, IMJ-PRG, Sorbonne Université, France
- 2022 **A5 seminar**, Instituto de Matemática e Estatística, Universidade de São Paulo, Brasil

#### To the general public

- 2021 **A Tale of Geometry**, *S4, Instituto de Matemática e Estatística*, Universidade de São Paulo, Brazil
- 2020 **Morse Theory in 15 Minutes**, *Senior Thesis Presentation, Instituto de Matemática e Estatística*, Universidade de São Paulo, Brazil
- 2020 **Is the Earth flat? A talk about curvature**, *S4, Instituto de Matemática e Estatística*, Universidade de São Paulo, Brazil
- 2020 **How to solve any equation, with some conditions**, *S4, Instituto de Matemática e Estatística*, Universidade de São Paulo, Brazil

### Other Info

#### Related Activities

- 2025 **Reading Seminar on Kähler Ricci Flow**, Online
- 2021 – 2022 **Reading Seminar on Wild Dynamics**, *Groupe De Travail Dynamiques Sauvages*, IMJ-PRG, Sorbonne Université

#### Languages

|            |                |
|------------|----------------|
| Portuguese | Native Speaker |
| Italian    | Native Speaker |
| English    | Fluent         |
| Spanish    | Fluent         |
| French     | Fluent         |
| Catalan    | Intermediate   |

#### Memberships

- 2026 Member at the Société Mathématique de France

#### Links

- ArXiv [https://arxiv.org/a/mesquitapiccione\\_p\\_1](https://arxiv.org/a/mesquitapiccione_p_1)
- Lattes <http://lattes.cnpq.br/7822745238118143>
- Orcid <https://orcid.org/0000-0002-1626-0726>